

TECHNICAL WHITE PAPER FOR HIGH-ASSURANCE SOFTWARE EVALUATION

Formally Constrained, Repository-Verified Bounded Encapsulation Interfaces

*Exact ABI State Transport, Constraint-Gated VM81 Admission,
and Native RNA Transcription in HHS Pass 219*

**Revision 3.0 - Expanded Computational and Mathematical Specification | Repository snapshot:
August 17, 2026**

Research relevance: high-assurance software architecture, deterministic state machines, exact ABI transport, and formally constrained execution.
Potential federal evaluation audiences include DARPA I2O and NIST ITL; no agency endorsement or procurement determination is implied.

REPOSITORY STATE AT PUBLICATION	
Canonical main	cc60b5741de32eb95566f7ba4977e7f1a15368ec - merge of frozen Pass 218 iterations 1-48; validated cumulative Pass 218 head bc8edd58f44da334781448272ae11165bfec681d.
Pass 219 1.11	Frozen feature-branch checkpoint b879214bbdedc90841642589a9db0e2878c0bbcc; implementation head b33a035468d0f130d3691c9e25261d25087caf72.
Canonical status	Pass 219 1.11 remains draft/unmerged. It is implementation evidence, not current main authority.
Post-freeze CI	Pass 219 RNA Rule Grammar ABI 1.11 run 32030375448: SUCCESS; Native RNA Transcription ABI 1.10 run 32030375245: SUCCESS.

This revision expands the supplied six-page draft and Revision 2.0 into a repository-aligned computational and mathematical specification; it does not modify or merge repository state.

Abstract

This revision aligns the original bounded-encapsulation white paper with the executable HHS repository rather than treating the six-page draft as the source of truth. The current exact ABI declares a fixed VM81 frame of 81 unsigned 64-bit words (5,184 bits / 648 bytes), eight ordered phase bases, 64 ordered phase pairs, exact little-endian frame import/export, and bounded x86_64 byte ingress/egress. Pass 219 then layers typed constraint admission, lossless Fibonacci composition, native RNA transcription views, and an executable RNA rule grammar over that inherited substrate. The result is a branch-scoped C11/C++17 implementation in which C++ value/view classes lower to stable C records and final canonical mutation remains in the inherited C VM81 authority.

The principal correction is one of claim scope. Compile-time layout constants and conformance tests establish specific ABI, ordering, range, rollback, and deterministic-transition properties; they do not by themselves prove absence of all memory corruption, side channels, undefined behavior, or cryptographic attack classes. Likewise, the prior draft's standalone zero-energy equation is not the governing repository ABI invariant. This paper instead reports the typed invariants actually present in Pass 219: exact Lo Shu numeral projections, distinct phase/metric projections, UQCEL admission masks, ordered xy/yx witnesses, 15,552-state coordinate bijection, Hash216 previous/change/receipt topology, and explicit RNA domain-rule state transitions.

1. Executive Summary

- The repository now supports a substantially broader system than the original paper described: the 648-byte VM81 transport frame is only the low-level carrier, not the complete Pass 219 architecture.
- Pass 218 is canonically merged on main through frozen iterations 1-48. Pass 219 1.11 is a validated, frozen, unmerged feature-branch checkpoint and must remain clearly distinguished from main.
- Pass 219 1.10 exposes ordered phase witnesses, trinary (xy, x+y, yx) gates, Hash72 token occurrences, a 216-position previous/change/receipt transition view, an inherited SHA-256 index resolver hook, and exact 15,552-state coordinate translation.
- Pass 219 1.11 adds fixed-capacity strand/domain/rule/program records, eight executable RNA rule kinds, C++ hhs::rna value classes, deterministic rule preconditions, transcription witnesses, and exact rollback to pre-program domain state.
- The no-float rule is enforced on the Pass 219 RNA delta by CI token scanning and by exact integer/BigUInt paths in the UQCEL admission profile. This is a scoped implementation property, not a claim that every historical or external workload contains no floating point.
- The correct assurance statement is therefore “repository-verified bounded interfaces and constrained state transitions,” with formal-method claims limited to properties that have an explicit proof artifact or exhaustive executable test.

1.1 Revision Map from the Supplied Draft

Draft topic	Repository-aligned revision
5184-bit cell wall	Retained as the exact VM81 frame constant: 81 x 64-bit words = 648 bytes. The repository exposes import/export APIs and ABI descriptors rather than a standalone C++ Frame5184 class.
Zero-drift rational proof	Replaced by executable typed invariants that are present in code/contract: exact rationals, phase ordering, Lo Shu numeral constraints, UQCEL masks, BigUInt canonical encodings, and no-float delta checks.
15,552 coordinate bijection	Retained and strengthened: the repository defines $u=243 \cdot o+g$ and exact inverse mapping to (trit, slot5184); 1.10 evidence reports exhaustive coverage including (63,242)→(2,5183).
C++ membrane class	Expanded into stable C ABI records plus C++ value/view wrappers for phase, hydration, Hash216 transition, RNA domain/program/rules, and witnesses.
Formal verification claim	Scoped to exact properties actually enforced by compile-time checks, strict compiler flags, deterministic tests, exhaustive finite-domain tests, and repository-visible receipts. Broader security remains a separate assurance task.
CI evidence	Updated from Pass 219 1.10-era evidence to frozen 1.11 and post-freeze success runs; current main is now the merged Pass 218 1-48 lineage.

2. Repository Evidence Basis

The evidence hierarchy used in this paper is intentionally repository-first:

1. Canonical main establishes inherited production history. At this snapshot, main is cc60b574... and records the merge of frozen Pass 218 iterations 1-48.
2. Frozen Pass 219 restart records establish bounded implementation status and explicitly distinguish validated branch work from canonical merge authority.
3. Header and implementation files define executable ABI semantics; contract amendments define required behavior but are not treated as proof of implementation by themselves.
4. CI workflows and conformance tests provide finite executable evidence for the scoped properties they actually exercise.
5. The supplied PDF is treated as the publication baseline to revise, not as authority when it conflicts with current repository state.

ASSURANCE CLASSIFICATION USED IN THIS PAPER

Implemented	Callable repository surface exists in the inspected branch.
Validated	A bounded test/CI record demonstrates the stated property on the identified commit.
Exhaustive finite test	All elements of an explicitly finite domain were exercised by the cited test.
Normative	Required by contract/amendment, but implementation must be separately evidenced.
Not established	The reviewed repository evidence does not justify the stronger claim.

3. Exact ABI Carrier and Bounded State Transport

The current exact ABI v1.1 defines the low-level carrier in plain C. The core constants are:

Repository ABI constants (hhs_runtime_exact_abi_v1_1_base.h)

```

HHS_EXACT_HASH72_LEN      = 72
HHS_EXACT_HASH72_COORDS   = 5184
HHS_EXACT_VM81_CELLS      = 81
HHS_EXACT_VM81_WORD_BITS  = 64
HHS_EXACT_VM81_FRAME_BITS = 5184
HHS_EXACT_VM81_FRAME_BYTES = 648
HHS_EXACT_PHASE_BASIS_COUNT = 8
HHS_EXACT_PHASE_PAIR_COUNT = 64

```

The frame type is structurally simple:

```

typedef struct HHSExactVM81Frame {
    uint64_t words[HHS_EXACT_VM81_CELLS];
} HHSExactVM81Frame;

```

The ABI also exports exact frame import/export functions, VM5184 address encode/decode, Hash72 coordinate encode/decode, ordered phase products, and bounded x86_64 instruction-byte ingress/egress. This is stronger repository evidence than the original illustrative C++ class because it identifies the actual public transport surface used by later Pass 219 layers.

3.1 What the 648-Byte Invariant Establishes

- A VM81 frame is exactly 81 64-bit words in the ABI definition.
- The declared frame size is 5,184 bits / 648 bytes and the ABI descriptor exposes those constants to callers.
- Little-endian byte import/export can be tested bit-for-bit at the 648-byte boundary.
- C++ wrapper records are required to remain standard-layout/trivially copyable where asserted, avoiding STL/vtable/allocator representation in the stable ABI.

3.2 What It Does Not Establish by Itself

- A fixed struct size does not prove that every caller is free of buffer overruns, use-after-free, data races, undefined behavior, compiler defects, DMA faults, or side-channel leakage.
- A static layout check does not prove that dynamic allocation is impossible elsewhere in the process.
- The repository evidence reviewed here is not a complete memory-safety proof for the full codebase or toolchain.

This qualification replaces the supplied draft's broader statement that layout assertions make memory overruns and dynamic allocation "provably impossible" at runtime.

4. Repository-Aligned Pass 219 Execution Stack

Pass 219 is not a replacement runtime. It is an additive organization and transcription layer over the inherited exact C authority. The current branch composition is summarized below so the execution boundary is visible without implying that the unmerged feature branch is canonical main.

Layer	Repository surface	Role / authority
Inherited exact ABI	VM81 frame; phase basis; Hash72/Hash216 transport; x86_64 ingress/egress	Canonical bounded carrier and mutation authority.
Pass 219 1.7	Lo Shu polynomial projection; typed phase and metric constraints	Exact symbolic/quantized constraint vocabulary.
Pass 219 1.8-1.9	UQCEL admission profile; Fibonacci descriptor composition	Constraint-gated composed admission and receipt formation.
Pass 219 1.10	Native RNA transcription C ABI + C++ value/view classes	Stable RNA-facing translation of inherited phase, transition, and hydration records.
Pass 219 1.11	Domain/strand/rule program; lineage; witness; rollback	Executable fixed-capacity RNA rule grammar with deterministic preconditions.
Canonical commit path	Inherited C VM81 admission/commit surface	Final state mutation remains outside the C++ organization layer.

Figure 1. Repository-aligned Pass 219 execution stack. The 1.11 feature branch is validated but unmerged; the stable C++ layer exposes and composes inherited exact state rather than redefining canonical mutation authority.

4.1 Ordered Native Phase Basis

The exact ABI registers eight ordered phase bases: x, y, z, w, xy, yx, zw, and wz. Pass 219 1.10 wraps these as PhaseOperator and OrderedPhaseProduct C++ value classes while preserving the underlying C witness record. Ordered source identity is therefore part of the state record rather than a display label.

Pass 219 trinary phase projection

```
T_phase = (xy, x+y, yx)

left  = xy
center = x+y
right = yx
```

Pass 219 1.10 exposes all three trinary identities and preserves xy/yx order through the native phase witness. The repository treats the center relation and the left/right ordered products as distinct fields rather than collapsing them into one scalar.

4.2 Hash72 and Hash216 Transition Topology

For one admitted transition, Pass 219 1.10 represents three ordered Hash72 lanes: previous state, change, and receipt. Their concatenation is a 216-character transition word. Every occurrence records its absolute position, lane role, lane-local position, glyph, and whether the inherited 32-byte SHA-256 index record has been resolved.

```
W216 = H_prev || H_change || H_receipt
      = 72 + 72 + 72 = 216 characters

positions 0..71    : PREVIOUS
positions 72..143  : CHANGE
positions 144..215 : RECEIPT
```

The 1.10 implementation deliberately uses an inherited index-resolver callback instead of inventing a new SHA-256 preimage or domain-separation rule. This preserves the distinction between "same glyph" and "same token occurrence": position, lane, transition identity, and predecessor/receipt ancestry remain part of occurrence identity.

5. Typed Exact Constraints Instead of a Standalone Zero-Energy Proof

The supplied draft centered a standalone $F(x,y,a,b)$ zero-energy derivation. That expression is not the governing executable Pass 219 ABI invariant in the inspected repository. This revision therefore reports the constraints that the repository actually formalizes and lowers.

5.1 Lo Shu Polynomial Numeral Surface

Pass 219 amendment 1.7 binds the exact projections $a^2=1$, $b^2=2$, $c^2=3$ to the Lo Shu polynomial surface:

```
L_H = {
  {b^4,      c^4,      b^2},
  {c^2,      b^2+c^2,  b^4+c^2},
  {b^6,      a^2,      b^2*c^2}
}

integer projection:
4 9 2
3 5 7
8 1 6
```

This is a typed numeral projection: the symbolic forms remain the native expressions, while the familiar 1..9 Lo Shu values are their exact integer projection for the declared constraints.

5.2 Distinct Phase and Quantization Projections

The repository explicitly prevents one common category error by separating u_phase from u_q . The cyclic phase-ring law and the dyadic quantization metric are registered as distinct projections:

Conventional scalar projections of the typed 1.7 constraints

```
u_phase^72 = 1
u_q^5256 * 2^66 = 1
```

The UQCEL admission path must not infer $u_q=1$ from phase closure or overwrite the u_phase relation. The typed constraint join carries compatible witnesses rather than forcing all views into one scalar identity.

5.3 Quadratic-Reciprocity Orientation

For the declared reciprocity profile, the repository maps the parity residue to ordered xy/yx orientation and to the two half-cycle phase addresses 0 and 36 modulo 72. This ordering is carried into the exact phase witness and tested as an algebraic relation independent of VM81 cell identity.

5.4 No-Float Authority Is Scoped and Enforced

The 1.11 CI workflow scans the new RNA rule delta for the tokens float and double and fails if either appears. The 1.8 UQCEL profile uses canonical BigUInt byte encodings and checked exact arithmetic for represented integer constraints. The resulting claim is deliberately scoped: the inspected Pass 219 authoritative delta rejects floating authority; this paper does not infer that every external dependency or every historical non-authoritative analysis contains no floating-point code.

6. Constraint-Gated Admission: Pass 219 1.8 and 1.9

6.1 UCE -> UQCEL -> VM81 Admission

Pass 219 1.8 promotes the Universal Constraint Envelope (UCE) into a finite exact admission profile rather than one untyped scalar equality. The implemented integer/symmetric profile validates source identity, canonical BigUInt encodings, Lo Shu/UQCEL constants, $P^2=p*q+\Delta$, A/B product constraints, reciprocity inputs, ordered xy/yx orientation, and a valid VM5184 address before a candidate frame can be committed.

```
native HARMONICCODE UCE
-> typed ConstraintJoin / projection selection
-> UQCEL exact witness
-> C ABI admission record
-> VM81 candidate admit / fail-closed reject
-> Hash72 change + receipt
-> Hash216 previous/change/receipt lineage
```

The full symbolic profile is registered but returns `UNSUPPORTED_DOMAIN` until residual symbolic clauses are lowered. That distinction is important for procurement-grade claim discipline: an implemented exact subprofile is not reported as evaluation of the unresolved symbolic domain.

6.2 Pass 192/216 Fibonacci Composition in the Canonical Pass 219 Path

Pass 219 1.9 repaired a composition gap by carrying the inherited Pass 192 nested Fibonacci schedule into the Pass 219 composed admission path. The documented schedule uses $F_0=1$, $F_1=2$, $F(n+2)=F(n+1)+F(n)$, exact ratios $F_n/F_{(n+1)}$, and a local membrane witness. For the current UCE delimiter depth of 10, the recorded witness is $F_{10}=144$, $F_{11}=233$, ratio 144/233, cumulative scale $1/144$, and membrane $10 \bmod 11 = 10$.

The 1.9 path binds the Fibonacci descriptor into final receipt material before the caller-visible VM81 commit. Tests require the composed receipt to differ from the bare UQCEL receipt for the same admitted input, making the inherited compression descriptor lineage-relevant rather than a sidecar.

7. Exact 15,552-State Coordinate Bijection and Hydration Views

The original paper's 15,552-state mapping is present in the current Pass 219 contract and 1.10 implementation, but the repository defines it precisely as a reversible change of coordinates between 64 operation states x 243 G243 states and 3 trits x 5,184 hydration slots.

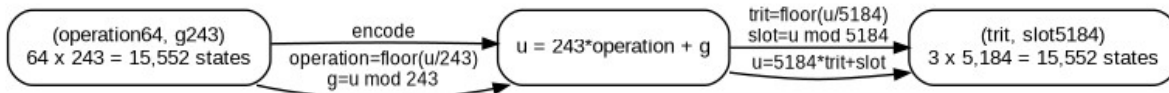


Figure 2. Exact local coordinate translation. The endpoint (operation=63, g=242) maps to (trist=2, slot=5183) and round-trips under the inverse formulas.

```

u = 243*operation64 + g243
trist = floor(u / 5184)
slot = u mod 5184

inverse:
u = 5184*trist + slot
operation64 = floor(u / 243)
g243 = u mod 243
  
```

The 1.10 restart record reports exhaustive testing of all $64 \times 243 = 15,552$ local states, uniqueness, inverse equality, and the operation-63 terminal coordinate. The same contract also preserves the first-level hydration cardinality in two exact factorizations:

```

81 * 41 * 64 * 243 = 51,648,192
81 * 3 * 5,184 * 41 = 51,648,192
  
```

These are coordinate decompositions of one declared fabric, not separate address authorities. VM81 cell, Lo Shu group, operation/G243 or trist/slot identity, ordered phase witness, and Hash72/Hash216 ancestry remain explicit where active.

8. Native RNA Transcription ABI - Pass 219 1.10

Pass 219 1.10 is the first repository surface in this paper that directly lowers the native transcription requirements into reusable C records and C++ value/views. The C API records include:

Record	Purpose
HHSExactPass219NativePhaseWitnessV1	Ordered left/right phase basis and ordered phase-product witness.
HHSExactPass219TrinaryPhaseGateV1	Trinary identity, xy and yx phase products, center relation, and source-order preservation.
HHSExactPass219Hash72TokenOccurrenceV1	Absolute 0..215 position, lane role, lane-local position, glyph, and 32-byte index record presence.
HHSExactPass219Hash216TransitionViewV1	Previous/change/receipt Hash72 strings, 216-character transition word/identity, 216 positional occurrence records.
HHSExactPass219HydrationCoordinateV1	Cell81, Lo Shu group, operation64, G243, trist, and slot5184 bridge.

Record	Purpose
HHSExactPass219RNAAdmissionV1	Composed admission aggregate combining native phase, trinary gate, hydration coordinate, inherited composed admission, and transition view.

The C++17 header wraps these records as PhaseOperator, OrderedPhaseProduct, TrinaryPhaseGate, Hash72TokenView, Hash216TransitionView, Hydration5184View, and RNAAdmissionView. The wrappers do not become mutation authorities; they expose and compose the stable C records.

8.1 1.10 Validation Boundary

- Strict C11 and C++17 compilation with -Wall -Wextra -Werror -pedantic.
- No new float/double authority in the 1.10 delta.
- Ordered $xy \neq yx$ identity and all three trinary identities.
- Exhaustive 15,552-state coordinate uniqueness and inverse equality, including operation 63 endpoint.
- Exact Hash216 3x72 lane offsets, all-or-nothing 216-entry inherited index resolver traversal, and 648-byte VM81 export/import round trip.
- Focused invalid/null/range rejection paths and inherited Pass 219 Fibonacci/composed-admission regression subset.

9. Executable RNA Rule Grammar - Pass 219 1.11

Pass 219 1.11 extends the 1.10 witness/transport layer into an executable fixed-capacity rule grammar. The C header defines at most eight domains per strand and sixteen rules per program. Each domain records a domain ID, complementary-domain ID, role flags, phase basis, and orientation. The program operates on a compact domain-state bitset with the following states: complement, bound, exposed, folded, active, inhibited, cleaved, and released.

9.1 Eight Rule Kinds and Deterministic Preconditions

Rule	State transition / precondition
Complement	Requires reciprocal complement-domain IDs and opposite orientation; marks both domains COMPLEMENT.
Binding	Requires both domains already marked COMPLEMENT; marks both BOUND.
Toehold	Requires source TOEHOLD role and source BOUND; marks source/target EXPOSED and clears FOLDED.
Hairpin	Requires source HAIRPIN role and source not EXPOSED; marks source FOLDED and clears ACTIVE.
Activation	Requires source BOUND and EXPOSED and not CLEAVED; sets ACTIVE and clears INHIBITED.
Inhibition	Sets INHIBITED and clears ACTIVE.
Cleavage	Requires source ACTIVE or INHIBITED; sets CLEAVED, clears source ACTIVE/BOUND, and clears target BOUND.
Release	Requires source CLEAVED; sets RELEASED and clears BOUND/FOLDED/EXPOSED.

Unmet rule preconditions return a constraint-rejected status rather than silently dispatching an unrelated operation. Program and domain capacities are checked, duplicate IDs are rejected, version/struct-size fields are validated, and domain phase basis/orientation/role flags are range-checked.

9.2 Stable C ABI + C++ hhs::rna Value Classes

The C++17 layer provides the classes Strand, Domain, Complement, Binding, ToeholdGate, HairpinGate, ActivationGate, InhibitionGate, Cleavage, Release, TranscriptionProgram, and TranscriptionWitness. They hold the C records by value and invoke the C ABI functions; standard-layout/trivially-copyable assertions are applied to the underlying ABI records.

C++17 reusable RNA rule classes

```

hhs::rna::Domain
hhs::rna::Strand
hhs::rna::Complement / Binding
hhs::rna::ToeholdGate / HairpinGate
hhs::rna::ActivationGate / InhibitionGate
hhs::rna::Cleavage / Release
hhs::rna::TranscriptionProgram
hhs::rna::TranscriptionWitness

```

9.3 Lineage and Rollback

A 1.11 transcription witness retains the 1.10 native phase witness, trinary gate, hydration coordinate, predecessor Hash72, and predecessor Hash216 identity. It also stores before/after domain state arrays and marks rollback availability. The conformance test executes a six-rule complement -> binding -> toehold -> activation -> cleavage -> release program, verifies final state flags, and then reconstructs the exact two-domain pre-program state through `hhs_exact_pass219_rna_witness_rollback`.

Separate tests confirm hairpin and inhibition behavior and explicitly require bare binding or release without prerequisites to return `CONSTRAINT_REJECTED`.

10. Verification and Reproducibility

PASS 219 1.11 FROZEN CHECKPOINT	
Branch	agent/pass219-iteration111-rna-rule-grammar-abi
Implementation head	b33a035468d0f130d3691c9e25261d25087caf72
Frozen head	b879214bbdedc90841642589a9db0e2878c0bbcc
Dedicated validation	run 32030254604 - SUCCESS at implementation head
Post-freeze branch run	run 32030375448 - SUCCESS at frozen branch head
Historical boundary	No Pass 212-218 deep scan, Genesis replay, broad unrelated regression, merge, or deployment was authorized by the 1.11 iteration.

The dedicated 1.11 workflow performs the following bounded gate:

1. reject float/double tokens in the 1.11 RNA rule delta
2. gcc -std=c11 -Wall -Wextra -Werror -pedantic exact ABI compile
3. run 1.11 C rule conformance test
4. g++ -std=c++17 -Wall -Wextra -Werror -pedantic C++ class conformance
5. run frozen 1.10 C admission regression
6. run frozen 1.10 C++ admission regression

On the frozen branch head, additional repository workflows also completed successfully, including the 1.10 Native RNA Transcription ABI workflow and the Pass 219 Universal Quantization Constraint Audit. Guarded Continuous Integration was skipped on that commit, so this paper does not silently equate “all repository workflows green” with the branch-scoped evidence above.

10.1 Current Main vs. Pass 219 Feature Branch

At publication, canonical main is the verified merge commit cc60b574... integrating frozen Pass 218 Iterations 1-48. The frozen Pass 219 1.11 branch remains unmerged and the repository compare reports divergence from current main. Therefore Pass 219 1.11 must be presented as validated implementation evidence, not as a deployed or canonical-main capability. A future merge must reconcile current main and rerun the appropriate exact-head/synthetic integration gates; this paper does not authorize or perform that operation.

11. Extended Computational and Mathematical Specification

Scope of this section. Sections 3-10 describe the directly inspected Pass 219 1.7-1.11 implementation and evidence boundary. This section expands the paper with the inherited HHS mathematics that Pass 219 is required to expose and compose. Where a relation is directly present in the frozen Pass 219 contract or implementation, it is identified as such. Where it belongs to an inherited Pass 217/218 design surface, it is labeled as inherited architecture rather than silently upgraded to a new 1.11 proof claim.

11.1 Literal RNA Transcription Over the Native Digital-DNA Algebra

Pass 219 amendment 1.5 closes the naming ambiguity around RNA and DNA. In this layer, the terminology denotes executable typed algebra over the native HHS phase substrate, not a decorative metaphor and not a conventional molecular simulator placed beside the runtime. The governing execution chain is:

Pass 219 native transcription authority chain

```
host x86_64 bytes / canonical memory carrier
-> exact C VM81 phase execution
-> ordered x,y,z,w noncommutative phase algebra
-> Hash72 external transition primitive
-> Hash216 previous/change/receipt vector surface
-> hydration / Lo Shu / qudit coordinate views
-> RNA typed rule composition
-> one inherited VM81 admission authority
```

For an admitted predecessor state S_n , the formal transcription program is represented as a typed transformation

```
Delta_RNA = T_RNA( $S_n$ , context, active_constraints, lineage)
```

The resulting candidate or delta is not canonical merely because it was produced by a C++ object. It must lower through the stable C ABI and the inherited VM81 admission surface. Operand order, reciprocal orientation, source grouping, predecessor state, and transition ancestry remain active parts of identity whenever the registered rule depends on them.

11.2 Ordered Phase Algebra and the 5,184-Address Operation Plane

The native ordered operator basis is $D=(x,y,z,w)$, with an exposed extension family that includes x, y, z, w, xy, yx, zw , and wz . HHS project specifications classify this as the reciprocal-phase/octonion-oriented digital-DNA substrate. The frozen Pass 219 ABI evidence directly preserves ordered products and phase witnesses; this paper does not invent a full Cayley multiplication table where the inspected ABI does not expose one. The Pass 219 rules preserve noncommutative source identity:

```
xy != yx
zw != wz
```

The exact ABI exposes eight phase bases. Ordered left/right selection therefore creates $8 \times 8 = 64$ phase-pair operations. Combined with 81 VM81 cells, the ordered address plane has exactly

```
81 cells x 8 left bases x 8 right bases = 81 x 64 = 5,184 addresses
```

This is the same cardinality as $72 \times 72 = 5,184$, but the coordinate meanings are not interchangeable by default. The 81 x 64 view carries VM81 cell and ordered operation identity; an inherited 72 x 72 view carries positional/transport identity. Pass 219 may translate between registered views only while retaining the witnesses required to reconstruct the original coordinate semantics.

11.3 Exact Numeral Basis and Lo Shu Polynomial Surface

The broader HHS exact numeral substrate uses symbolic square constraints rather than floating approximations. The active Pass 219 1.7 Lo Shu profile requires $a^2=1$, $b^2=2$, and $c^2=3$. The inherited project basis also registers $d^2=5$, with optional higher Fibonacci-associated lifts $e^2=8$, $f^2=13$, and $g^2=21$ where the corresponding pass contract activates them. Pass 219 1.7 itself only depends on the a,b,c subset for the equations below.

Exact Lo Shu polynomial numeral surface

```

a^2 = 1
b^2 = 2
c^2 = 3

L_H = {
  {b^4,          c^4,          b^2},
  {c^2,          b^2+c^2,      b^4+c^2},
  {b^6,          a^2,          b^2*c^2}
}

integer projection:
4 9 2
3 5 7
8 1 6

```

Each row, column, and main diagonal projects to the exact magic sum 15. The complete 3 x 3 projection sums to 45. The amendment also defines fixed symbolic numeral lifts so downstream equations can refer to exact polynomial identities rather than decimal literals:

Lift	Native expression	Integer projection
N1	a^2	1
N2	b^2	2
N3	c^2	3
N4	b^4	4
N5	b^2+c^2	5
N6	b^2*c^2	6
N7	b^4+c^2	7
N8	b^6	8
N9	c^4	9
N12	c^2*b^4	12
N36	b^4*c^4	36
N72	b^6*c^4	72
N73	$b^6*c^4+a^2$	73
N66	$b^6*c^4-b^2*c^2$	66
N5256	$(b^6*c^4)*(b^6*c^4+a^2)$	5256
ZERO_L	c^2-c^2	0

11.4 Two Typed u Projections: Phase Closure and Dyadic Quantization

Pass 219 1.7 explicitly prevents scalar conflation by separating the cyclic phase projection u_{phase} from the exact dyadic quantization projection u_q . The phase law is

```

u_phase^(b^6*c^4) = a^2
conventional projection: u_phase^72 = 1

```

The quantization metric begins from the exact symbolic relation

```

(b^2)^(a^2 / ((c^2*b^4)*pi_scalar(xy)))
=
b^2 * u_q^(b^6*c^4+a^2)

```

For the declared unit-product scalar projection $\text{pi_scalar}(xy)=a^2$, substituting the exact numeral lifts yields

Exact single-step dyadic metric derivation

```

2^(1/12) = 2 * u_q^73
u_q^73 = 2^(1/12 - 1)
        = 2^(-11/12)

```

Applying one complete N72 cycle multiplies both exponents by 72. Therefore $73 \times 72 = 5,256$ and $(-11/12) \times 72 = -66$, giving the exact full-cycle closure

Pass 219 1.7 full-cycle quantization closure

```
u_q^5256 = 2^-66
u_q^5256 * 2^66 = 1
```

The equality above is a typed metric constraint; it does not imply $u_q=1$ and does not replace $u_{\text{phase}}^{72}=1$. Both projections may belong to one admitted source state only through an explicit projection record and compatible reconstruction witness.

11.5 Quadratic-Reciprocity Orientation as an Ordered Phase Witness

For positive odd reciprocity inputs p and q , amendment 1.7 defines the exact Lo Shu parity residue

```
epsilon_L(p,q) = Mod(((p-a^2)*(q-a^2))/b^4, b^2)
with a^2=1, b^4=4, b^2=2:
epsilon_L(p,q) = ((p-1)(q-1)/4) mod 2
```

The ordered reciprocal lane is then selected by

```
epsilon_L = 0  -> xy
epsilon_L = 1  -> yx

Phi_QR(p,q) = Mod((b^4*c^4)*epsilon_L, b^6*c^4)
              = 36*epsilon_L mod 72
```

Thus xy maps to phase address $0 \bmod 72$ and yx maps to the half-cycle phase address $36 \bmod 72$. For odd primes, the yx branch is selected precisely when both inputs occupy the $3 \bmod 4$ residue class. The scalar sign pair $(+1,-1)$ is only a compatibility projection; the native identity is the ordered xy/yx provenance.

11.6 UCE -> UQCEL Exact Admission Mathematics

Pass 219 1.8 promotes the typed Universal Constraint Envelope into the first enforceable finite exact profile. The source fixture is bound by ASCII-normalized SHA-256 `7eb0cc5707a4a58a5a8e4879e0e2e3bdab22c15fe4503fb3a3b0e16596343d42`. The integer/symmetric profile admits only when all represented constraints are simultaneously satisfied.

```
P^2 = p*q + Delta
A = P^2
B = P^2
A*B = P^4
p,q are positive odd reciprocity inputs
QR parity selects the expected xy/yx lane
observed ordered phase equals 0 or 36 as required
VM5184 address is valid
Lo Shu and UQCEL constants are exact
BigUInt encodings are canonical
```

The input integers use canonical minimal big-endian BigUInt views inherited from the Pass 133/211 BigInt serialization authority. This is distinct from the 648-byte VM81 frame transport, whose byte import/export contract is little-endian. The two byte orders serve different typed interfaces and must not be conflated.

The full symbolic profile is registered but fails closed with `UNSUPPORTED_DOMAIN` while the residual `T_M_HARMONIC`, `Tensor_S_F_AT_BT`, `DELTA_P_ROOT`, and `MOD_F_U` clauses remain unlowered. An unresolved clause is not treated as false and cannot authorize a full-symbolic commit.

11.7 Fail-Closed Commit Semantics and Exact Negative Behavior

The UQCEL gate zeros the output commit frame before evaluation. Only an `ADMIT` decision may copy the candidate 648-byte frame into committed output. `REJECT` and `UNSUPPORTED_DOMAIN` leave the commit frame zero. No rejection may be repaired by approximate arithmetic.

Rejected condition	Required behavior
Source fixture hash mismatch	Fail closed; no canonical commit.
Non-canonical BigUInt encoding	Fail closed; no reinterpretation through host numeric types.
$P^2 \neq p*q + \Delta$	Constraint rejection.

Rejected condition	Required behavior
$A \neq P^2$, $B \neq P^2$, or $A*B \neq P^4$	Constraint rejection.
Even/nonpositive reciprocity input	Constraint rejection.
xy/yx orientation mismatch	Constraint rejection; phase ancestry preserved in receipt where structurally valid.
Invalid VM5184 address or basis	Range/constraint rejection.
Full-symbolic residual clauses unresolved	UNSUPPORTED_DOMAIN.
Checked-arithmetic overflow/bounds failure	Fail closed; never approximate.

11.8 Hash72 Primitive Language and Hash216 Positional Vector Indexing

The native external transition language is positional. For one Hash72 record

```
H72 = (g_0, g_1, ..., g_71)
```

each occurrence preserves glyph identity, position 0..71, source VM81 transition identity, ordered phase ancestry, lane role, and receipt lineage. The complete transition combines three ordered records:

```
H_prev    = previous-state Hash72
H_change  = current state-change Hash72
H_receipt = execution/closure receipt Hash72
```

```
W216 = H_prev || H_change || H_receipt
|W216| = 216
```

For each i in 0..215, the inherited Hash216 schema resolves one domain-separated positional SHA-256 index record. Pass 219 1.10 intentionally calls the inherited resolver rather than redefining its byte preimage:

```
V216[i] = SHA256_INDEX_RECORD(
    transition_identity,
    lane_role,
    position,
    W216[i]
)
```

The exact byte preimage and domain separator remain inherited implementation authority. A useful derived size fact is that 216 independent 32-byte SHA-256 records contain 6,912 raw digest bytes (55,296 bits) if materialized contiguously; this arithmetic is a vector-surface size observation, not a statement that the stable Pass 219 ABI stores them as one 6,912-byte struct. The crucial invariant is positional identity: the same glyph at two different lanes, positions, or predecessor states is not the same token occurrence.

11.9 Hydration Coordinate Algebra and Exact Cardinality Factorizations

The Pass 219 transcription coordinate system is built from equal cardinalities rather than heuristic embedding dimensions. Three identities are central:

```
81 * 64 = 5,184
72 * 72 = 5,184
3 * 5,184 = 64 * 243 = 15,552
```

The exact local bijection is

```
u = 243*operation64 + g243
trit = floor(u / 5184)
slot5184 = u mod 5184

inverse:
u = 5184*trit + slot5184
operation64 = floor(u / 243)
g243 = u mod 243
```

With 81 VM81 cells and 41 Lo Shu groups, the first-level contextual fabric has two exact decompositions:

```
81 * 41 * 64 * 243 = 51,648,192
81 * 3 * 5,184 * 41 = 51,648,192
```

The tuple (cell81, lo_shu_group41, operation64, g243) therefore round-trips to (cell81, lo_shu_group41, trinary_gate, hydration5184_slot) without creating a second state universe. The coordinate change reorganizes the same finite set.

Pass 219 1.10 exposes the bridge as HHSExactPass219HydrationCoordinateV1 and exhaustively tests the 15,552-state local component.

11.10 Inherited 81-Cell Lo Shu/Sudoku Qudit and Genesis Geometry

The inherited Pass 217 architecture treats the 81-cell substrate as a nested exact Lo Shu/Sudoku qudit rather than an unstructured array. The local 3 x 3 Lo Shu invariant supplies a reversible discrete constraint surface; the broader 9 x 9 / 81-cell construction reuses those local relations, while the 41-group coordinate provides an inherited contextual grouping axis. Pass 219 1.10 exposes the group coordinate but does not independently re-prove the entire Pass 217 nested geometry.

In the inherited Genesis-state design, the central Lo Shu relation is preserved while outer positions may carry the eight ordered phase channels x, y, z, w, xy, yx, zw, wz. Row, column, diagonal, and wraparound traversals are therefore not merely spatial presentation choices: they are candidate closed paths through exact phase and cell identities. This geometry is an inherited state-organization specification that Pass 219 must preserve when a transcription rule touches those coordinates.

Inherited architecture note: the complete nested Sudoku/Lo Shu and Genesis construction belongs to earlier pass contracts. The current Pass 219 1.10/1.11 evidence proves the exposed coordinate and RNA interfaces, not every theorem of the inherited geometry.

11.11 Exact BigInt, Rational, and x86_64 Serialization Boundaries

Canonical HHS authority is exact. Pass 219 therefore separates three representation responsibilities: symbolic/rational value identity, arbitrary-precision integer transport for values exceeding one host word, and the fixed 648-byte VM81 frame for x86_64-aligned state transport. No floating-point conversion may become an authoritative bridge between them.

Surface	Exact representation requirement	Pass 219 implication
UQCEL integer inputs	Canonical minimal big-endian BigInt views	Checked byte arithmetic; no narrowing to float or one machine word.
Exact rational/symbolic constraints	Reduced rational / symbolic exponent / polynomial identity	Keep u_phase, u_q, Lo Shu, reciprocity, and source witnesses typed.
VM81 frame	81 uint64_t words; exact 648-byte little-endian import/export	Bit-for-bit transport and replay boundary.
C++ organization layer	Value/view wrappers over stable C records	No STL/vtable/allocator representation crosses the stable public ABI.
Compact/hydrated state	Exact reconstruction witness bound to compact carrier	Compression may reduce representation, never discard phase/lineage required for reconstruction.

11.12 Deterministic Contraction, Hydration, and Reverse Reconstruction

The inherited architecture treats hydration as the inverse of deterministic contraction, not as a probabilistic approximation step. Three round-trip laws summarize the required behavior over their declared domains:

```

DECOMPOSE_TRINARY(RECONSTRUCT_TRINARY(witness)) = witness
CONTRACT(HYDRATE(S)) = S
DECOMPRESS(COMPRESS(S)) = S

```

Lossless compression is therefore modeled as expanded deterministic state -> canonical generator/compact state -> exact exceptions where required -> authenticated lineage/reconstruction witness. Hydration applies the inherited rules and exact exceptions to recover the same canonical serialization. Pass 219 consumes previously validated contraction/hydration evidence; it does not obtain permission to quote a new compression ratio for a different workload solely because the geometry is shared.

11.13 Pass 218 Continuation Equivalence as the Pass 219 Activation Premise

The 1.5 amendment explicitly makes Pass 219 an additive transcription layer over the already-validated continuation system. Its activation premise is that the full first-principles/Genesis reference path and the optimized indexed-continuation path terminate at the same canonical serialization and cryptographic identity for the declared Pass 218 domain:

```

full first-principles / Genesis reference path
-> canonical target serialization
-> SHA-256 X

optimized indexed continuation / hydration path
-> same canonical target serialization
-> SHA-256 X

```

The practical consequence for Pass 219 is compositional: RNA rules start from Hash216-addressed, already-admitted state and transcribe only the required delta. Cache reuse, branch prediction, hydration ROM lookup, parallel scheduling, or GPU preparation may accelerate retrieval or candidate construction, but no optimized path may become a second canonical state authority. Equality is judged at the same exact serialized/receipt boundary.

11.14 C and C++ Capability Mapping

The long-form 1.5 contract describes the target reusable C++ capability vocabulary. The frozen 1.10/1.11 branch implements a substantial subset as C++17 value/view classes over stable C records. The distinction between target vocabulary and current branch implementation is shown below.

Capability family	Contract vocabulary	Frozen 1.10/1.11 implementation status
Phase	PhaseOperator, OrderedPhaseProduct, ReciprocalPhaseRelation, TrinaryPhaseGate	PhaseOperator, OrderedPhaseProduct, TrinaryPhaseGate implemented; ordered witnesses lower to C records.
Transition	VM81StateView, VM81Delta, Hash72Primitive, Hash72TokenView, Hash216TransitionVector	Hash72TokenView and Hash216TransitionView implemented; VM81 authority remains inherited C runtime.
Hydration	LoShuCell, LoShuGroup41, Qudit81View, Hydration5184View, HydrationROM51648192View	Hydration5184View and coordinate record implemented; deeper ROM/Lo Shu classes remain inherited/target vocabulary rather than all separately instantiated in 1.10.
Exact transport	ExactBigIntState, CanonicalByteView	Exact BigInt/UQCEL and fixed VM81 byte transport inherited/used; no floating authority.
RNA	Strand, Domain, Complement, Binding, ToeholdGate, HairpinGate, ActivationGate, InhibitionGate, Cleavage, Release, TranscriptionProgram, TranscriptionWitness	Implemented in 1.11 with fixed-capacity records, deterministic preconditions, lineage and rollback.

11.15 Stable ABI Lowering Rules

Any Pass 219 organization class capable of producing an authoritative successor candidate must lower to C-compatible records. The stable public ABI may carry or reference predecessor VM81 identity, predecessor Hash72/Hash216 identity, ordered phase operands, trinary decomposition witness, active 81/5,184/41-group coordinates, exact BigInt/canonical byte references, dependency frontier, program identity, candidate delta, and rollback witness. The public ABI must not depend on implementation-specific STL layout, vtables, exception objects, allocator pointers, or C++ object-model details.

The current frozen branch validates this model using C11 and C++17. Broader Pass 219 design text may describe later C++20 compound-constraint organization, but that target does not change the frozen 1.10/1.11 evidence language: final mutation remains in the inherited C VM81 authority.

11.16 RNA State Machine as a Deterministic Algebraic Program

The eight RNA rule kinds can be read as a deterministic state-transition algebra over domain-state bits. One validated conformance path is:

```

COMPLEMENT
-> BINDING
-> TOEHOLD EXPOSURE
-> ACTIVATION
-> CLEAVAGE
-> RELEASE

```

The rule interpreter records before-state and after-state arrays in the transcription witness. Because rollback restores the exact before-state, the grammar is not merely a forward dispatch list: it exposes a reversible execution boundary for the finite domain-state component. Hairpin and inhibition form additional branches, and invalid bare binding/release attempts are rejected by precondition checks rather than coerced into a valid state.

11.17 Hash216-Indexed Continuation, Vector Cache, and Acceleration Invariants

Pass 219 is designed to organize existing deterministic acceleration rather than bypass it. The inherited optimization layers use exact transition identity and hydration lineage to avoid recomputing already-proven state. Their authority boundary is semantic: an optimization may change how a candidate is found or scheduled, but it may not change the exact canonical result.

Optimization surface	Computational role	Required invariant
Hash216 identity/reuse lookup	Resolve previously admitted transition/token occurrences and continuation candidates from the indexed vector surface.	Lookup result remains bound to predecessor, lane, position, receipt, and exact state identity.
Vector cache	Retain hydrated or compact exact state for repeated continuation.	Cache hit is a reuse optimization, not a second source of truth; serialized state must equal canonical storage.
Delta continuation	Retrieve the admitted predecessor and evaluate only the active dependency frontier / changed state.	Result must equal the corresponding deterministic full execution at the canonical serialization/receipt boundary.
Cache eviction + rehydration	Discard materialized expansion and later reconstruct from compact state, exact exceptions, and inherited rules.	Rehydrated state must round-trip bit-for-bit to the same admitted state.
Branch prediction / scheduling	Prioritize likely continuation branches or reorder candidate work.	Prediction is advisory; no predicted branch commits without the normal exact admission checks.
Parallel / GPU preparation	Evaluate independent candidate work or prepare branch manifolds concurrently.	Parallelism may not introduce an alternative mutation authority or float-based canonical decision path.

This is the computational meaning of “retrieve proven state before transcribing only the needed delta”: Pass 219 should compose already-validated Hash216/vector/hydration machinery first, then lower only the new RNA-organizational change through the same VM81 admission authority.

11.18 Auxiliary 64-State Hexagram / Hash216 Lattice Mapping

A recent auxiliary HHS lattice module organizes 64 hexagram states as an 8 x 8 ordered pair of trigrams and maps that finite set into the three Hash216 lane bands. This module is useful for showing how a 64-state symbolic organization can reuse the same positional transition surface without redefining Hash216.

Auxiliary 64-state / three-band mapping

```

TRIGRAMS: T1..T8
mirror pairs: T1<->T5, T2<->T6, T3<->T7, T4<->T8

H[upper,lower] with upper,lower in 0..7
idx64 = 8*upper + lower
0 <= idx64 <= 63

band offsets = {0, 72, 144}
hash216_address = (idx64 + band_offset) mod 216

```

The resulting address ranges are 0..63 for the previous lane band, 72..135 for the change lane band, and 144..207 for the receipt lane band. The remaining eight positions in each 72-position lane remain outside this 64-state adapter, which is consistent with the fact that the adapter is a 64-state projection into a 72-position primitive language rather than a replacement for the full Hash72 lane.

The same module registers the seed basis (x,y,xy,yx), the additive relation $\text{Sigma} = x+y = y+x = xy+yx = yx+xy$, and $Q = x^2y^2 + \text{Sigma}$. These equalities concern the registered additive/composite projection; they do not erase the ordered-product witness $xy \neq yx$. This auxiliary mapping should be evaluated as a separate integration surface unless and until its repository checkpoint is explicitly incorporated into the frozen Pass 219 branch.

11.19 Mathematical and Computational Conformance Checklist

The 1.7 amendment defines a precise set of conformance obligations for the quantization correspondence. They are useful as a compact mathematical checklist for external evaluators:

ID	Required property
P219-UQ01	Lo Shu polynomial projects exactly to $\{\{4,9,2\},\{3,5,7\},\{8,1,6\}\}$.
P219-UQ02	Every Lo Shu row, column, and diagonal has the same exact magic sum.

ID	Required property
P219-UQ03	N12, N36, N72, N73, N66, N5256 project exactly to 12,36,72,73,66,5256.
P219-UQ04	Unit-product metric projection derives exponent -11/12 before the full cycle.
P219-UQ05	Full N72 cycle derives the exact base- b^2 exponent -66.
P219-UQ06	Polynomial metric closure projects to $u_q^{5256} \cdot 2^{66} = 1$.
P219-UQ07	ϵ_L is ZERO_L except when both odd residue classes are 3 mod 4.
P219-UQ08	$\epsilon_L = a^2$ selects $yx/\text{phase } N36$; ZERO_L selects $xy/\text{phase ZERO}_L$.
P219-UQ09	Exact ABI $x \cdot y$ returns $xy/\text{phase ZERO}_L$; $y \cdot x$ returns $yx/\text{phase } N36$.
P219-UQ10	xy/yx ordered witness is preserved across all 81 VM81 cells.
P219-UQ11	All 5,184 VM81 ordered phase addresses round-trip exactly.
P219-UQ12	Every exact ABI phase product remains in the inherited quarter-cycle phase set.
P219-UQ13	New UQCEL reference/oracle source contains no float/double/transcendental canonical path.
P219-UQ14	u_{phase} closure and u_q metric projection remain type-distinct.
P219-UQ15	Substrate conformance is not mislabeled mandatory admission enforcement before the gate exists.

11.20 Consolidated Cardinality and Representation Table

Quantity	Exact value	Meaning
Hash72 length	72	Primitive external transition positions per lane.
Hash216 length	$216 = 3 \times 72$	Previous/change/receipt transition word.
Phase basis count	8	$x, y, z, w, xy, yx, zw, wz$.
Ordered phase pairs	$64 = 8 \times 8$	Operation identity per VM81 cell.
VM81 cells	81	Fixed 64-bit words / cell identities.
VM81 frame	5,184 bits = 648 bytes	81 x 64-bit exact transport carrier.
VM5184 address plane	$5,184 = 81 \times 64$	Cell x ordered operation.
Alternate 5,184 geometry	$5,184 = 72 \times 72$	Inherited positional/transport factorization.
G243 coordinate	243	Inherited local G243 state count.
Local transcription states	$15,552 = 64 \times 243 = 3 \times 5,184$	Exact operation/G243 $\leftrightarrow$ trit/hydration-slot bijection.
Lo Shu groups	41	Inherited contextual grouping coordinate exposed in hydration view.
First-level hydration ROM	51,648,192	$81 \times 41 \times 64 \times 243 = 81 \times 3 \times 5,184 \times 41$.
Raw 216 SHA-256 records	6,912 bytes if contiguous	Derived 216 x 32-byte positional digest materialization; not a stable-ABI struct-size claim.

12. Claim/Evidence Matrix

Claim	Evidence status	Repository basis
VM81 frame = $81 \times 64\text{-bit} = 648$ bytes	Implemented	Exact ABI constants and HHSExactVM81Frame definition.
Exact 648-byte import/export round trip	Validated	1.10 focused conformance evidence.
$64 \times 243 \leftrightarrow 3 \times 5184$ bijection	Exhaustive finite test	1.10 reports all 15,552 states plus operation-63 endpoint.
xy/yx ordered identities preserved	Validated	Exact phase witness + 1.10 trinary tests + 1.11 lineage reuse.

Claim	Evidence status	Repository basis
Hash216 has ordered previous/change/receipt topology	Implemented + validated	1.10 transition record and lane-offset tests.
Every Hash216 index uses inherited positional SHA-256 schema	Interface preserved	1.10 resolver hook preserves inherited schema; this revision does not re-prove the historical SHA-256 preimage/domain-separation implementation.
UQCEL integer/symmetric profile gates commit	Implemented	1.8 exact admission profile; full symbolic profile explicitly unsupported.
Fibonacci descriptor participates in composed receipt	Implemented/tested	1.9 composed admission and receipt-difference evidence.
Eight RNA rule kinds execute with preconditions	Validated	1.11 C implementation and conformance tests.
RNA witness exact rollback	Validated	1.11 rollback API and test restoring before-state.
No float/double authority in 1.11 delta	Validated	Dedicated CI lexical rejection step; 1.10 has equivalent scoped check.
All memory corruption impossible	Not established	Fixed ABI size does not constitute a whole-program memory-safety proof.
Post-quantum security proven by this pass	Not established by reviewed Pass 219 evidence	Requires separate cryptographic construction, threat model, and proof/certification evidence.
Pass 219 1.11 is canonical main	False at snapshot	Frozen branch is explicitly DRAFT/UNMERGED; main is the Pass 218 merge head.

13. High-Assurance Evaluation Roadmap

The repository is now suitable for a more rigorous evaluation package than the supplied draft described, but the next assurance steps should be explicit about which layer is being certified.

1. Reconcile the frozen Pass 219 1.11 branch with current main under the project's guarded merge policy and preserve exact commit/evidence lineage. This is repository integration work, not performed by this document revision.
2. Re-run the dependency-scoped C/C++ ABI gates and a synthetic merge gate against the exact merge candidate; record artifact hashes and compiler/toolchain versions.
3. Add machine-checkable layout assertions for public ABI structs where size/alignment is contract-relevant, while keeping semantic invariants in runtime tests rather than overloading sizeof assertions.
4. For claims labeled 'formal verification,' identify the exact property and proof mechanism (e.g., exhaustive finite enumeration, SMT/CBMC/Frama-C proof, model checking, or theorem prover) rather than using the term as a blanket synonym for deterministic testing.
5. Define an explicit threat model for memory corruption, concurrency, compiler/toolchain trust, side channels, malicious input, and cryptographic index integrity before making security claims beyond deterministic state/ABI behavior.
6. Map the resulting evidence package to the specific requirements of any DARPA/NIST/NSA program or standard only after verifying the current solicitation/standard text. The repository itself does not establish agency endorsement.

14. Reproducibility Appendix

ID	Repository evidence
R1	main @ cc60b5741de32eb95566f7ba4977e7f1a15368ec - "Merge frozen Pass 218 Iterations 1-48"; cumulative frozen Pass 218 head bc8edd58f44da334781448272ae11165bfec681d.
R2	docs/operations/restart/ PASS_219_NATIVE_RNA_TRANSCRIPTION_ABI_1_10_RESTART.md
R3	docs/operations/restart/ PASS_219_RNA_RULE_GRAMMAR_ABI_1_11_RESTART.md at frozen branch head b879214...

ID	Repository evidence
R4	hhs_runtime/include/hhs_runtime_exact_abi_v1_1_base.h
R5	hhs_runtime/include/hhs_pass219_rna_transcription_1_10.h and .hpp
R6	hhs_runtime/include/hhs_pass219_rna_rule_grammar_1_11.h and .hpp; hhs_runtime/c/hhs_pass219_rna_rule_grammar_1_11.inc
R7	tests/pass219/test_pass219_rna_rule_grammar_1_11.c and .cpp
R8	HHS_PASS_219_APPEND_ONLY_NATIVE_RNA_TRANSCRIPTION_ABI_AMENDMENT_1_5_0.md
R9	HHS_PASS_219_APPEND_ONLY_LO_SHU_DYADIC_QUADRATIC_RECIROCITY_QUANTIZATION_AMENDMENT_1_7_0.md
R10	HHS_PASS_219_APPEND_ONLY_NATIVE_UNIVERSAL_CONSTRAINT_ENFORCEMENT_AMENDMENT_1_8_0.md
R11	docs/operations/restart/ PASS_219_NESTED_MODULAR_FIBONACCI_COMPRESSION_1_9_RESTART.md
R12	.github/workflows/pass219-rna-rule-grammar-1-11.yml; branch run 32030375448 SUCCESS.
B0	Supplied six-page white paper dated August 17, 2026: "Formally Verified Bounded Encapsulation Interfaces: Exact Rational Arithmetic and Deterministic Memory Layouts in High-Consequence Systems."

14.1 Exact Commands Represented by the 1.11 Gate

```
gcc -std=c11 -Wall -Wextra -Werror -pedantic \
-Ihhs_runtime/include -c hhs_runtime/c/hhs_runtime_exact_abi.c

gcc -std=c11 -Wall -Wextra -Werror -pedantic \
-Ihhs_runtime/include tests/pass219/test_pass219_rna_rule_grammar_1_11.c ...

g++ -std=c++17 -Wall -Wextra -Werror -pedantic \
-Ihhs_runtime/include tests/pass219/test_pass219_rna_rule_grammar_1_11.cpp ...
```

15. Conclusion

The expanded computational and mathematical specification supports a materially richer and more defensible white-paper claim than the original six-page draft: HHS exposes an exact 648-byte VM81 transport ABI, ordered noncommutative phase witnesses, fixed Hash72/Hash216 lineage, an exhaustively tested 15,552-state coordinate translation, constraint-gated UQCEL admission, inherited Fibonacci composition, native RNA transcription views, and an executable RNA rule grammar with deterministic preconditions and rollback. These properties are connected through stable C records and C++ value/view classes without transferring canonical mutation authority out of the inherited C VM81 runtime.

The strongest version of the paper is also the most precise. The current evidence supports deterministic, bounded, exact, repository-reproducible interface claims on identified commits and finite domains. It does not yet justify whole-program memory-safety, universal formal verification, post-quantum security, or canonical-main Pass 219 completion. Maintaining that boundary makes the document suitable for serious technical evaluation because every major assertion is tied either to executable repository evidence or to a clearly labeled future assurance requirement.